ESKWELA Database Documentation & Technical Specification

Role: Database Manager & Engineer | **Backend Architecture:** Google Firebase Auth & Firestore / Structured Relational Export

1. Executive Summary

This document provides the complete database technical specifications, instructor-formatted Data Dictionary, field-precise visual ERD diagram, role-restricted Use Case Diagram, DFD, Sequence Diagram, and raw DDL SQL creation script for **ESKWELA**. Administrators have restricted privilege scoped exclusively to the Admin Control Center, user inspection, and modifying announcements.

2. Data Dictionary (Instructor Format)

tblUsers (Users Collection)

Field Name	Table	Database Field Name	Data Type	Length/Decimal Places	Description
User ID	tblUsers	id	Text	128	Firebase Auth UID / Primary Key
Email Address	tblUsers	email	Text	255	Unique student or admin login email
Full Name	tblUsers	fullName	Text	150	Student or administrator full name
Student ID	tblUsers	studentId	Text	50	Unique student ID card number
School Name	tblUsers	school	Text	200	University or school name
Course Program	tblUsers	course	Text	150	Enrolled degree program
Year Level	tblUsers	year	Text	20	Academic year level (e.g. 4th Year)
Class Section	tblUsers	section	Text	50	Class section code (e.g. IT41C1)
Role Privilege	tblUsers	role	Text	30	Access role: 'student' or 'admin'
Account Status	tblUsers	status	Text	30	Status: active, suspended, deactivated, archived
Suspension Duration	tblUsers	suspensionDuration	Text	50	Duration if suspended (e.g. 24 Hours)
Suspended Timestamp	tblUsers	suspendedAt	Number	-	Epoch timestamp when account was suspended

tblSchedules (Schedules Collection)

Field Name	Table	Database Field Name	Data Type	Length/Decimal Places	Description
Schedule ID	tblSchedules	id	Text	128	Unique schedule record Primary Key
User ID	tblSchedules	userId	Text	128	Foreign Key referencing tblUsers(id)
Schedule Date	tblSchedules	dateString	Text	20	ISO Scheduled date string (YYYY-MM-DD)
Class Name	tblSchedules	className	Text	200	Course title (e.g. CAPSTONE PROJECT 1)

Class Code	tblSchedules	classCode	Text	50	Course code (e.g. IT6398)
School Name	tblSchedules	school	Text	150	Institution name
Start Time	tblSchedules	startTime	Text	30	Class start time (e.g. 3:00 PM)
End Time	tblSchedules	endTime	Text	30	Class end time (e.g. 4:30 PM)
Room Location	tblSchedules	room	Text	50	Room or laboratory number (e.g. 301, SLAB5)
Instructor / Units	tblSchedules	instructor	Text	150	Instructor name or section/units metadata
Color Badge	tblSchedules	colorType	Text	30	UI color badge (BLUE, ORANGE, GREEN)

tblTasks (Tasks Collection)

Field Name	Table	Database Field Name	Data Type	Length/Decimal Places	Description
Task ID	tblTasks	id	Text	128	Unique task record Primary Key
User ID	tblTasks	userId	Text	128	Foreign Key referencing tblUsers(id)
Schedule ID	tblTasks	scheduleId	Text	128	Optional Foreign Key referencing tblSchedules(id)
Task Title	tblTasks	title	Text	500	Task description or assignment title
Due Date	tblTasks	dateString	Text	100	Deadline display string
Course Name	tblTasks	courseName	Text	150	Associated course name
Task Priority	tblTasks	priority	Text	30	Priority level: HIGH, MEDIUM, LOW
Completion Status	tblTasks	completed	Boolean	-	Completion flag: true or false

tblAiQuizzes (AI Quizzes Collection)

Field Name	Table	Database Field Name	Data Type	Length/Decimal Places	Description
Quiz ID	tblAiQuizzes	id	Text	128	Unique AI quiz record Primary Key
User ID	tblAiQuizzes	userId	Text	128	Foreign Key referencing tblUsers(id)
User Email	tblAiQuizzes	userEmail	Text	255	Email prefix used for Leaderboard ranking
Quiz Topic	tblAiQuizzes	topic	Text	250	Quiz course topic or subject
Correct Score	tblAiQuizzes	score	Number	-	Number of correct answers
Total Questions	tblAiQuizzes	totalQuestions	Number	-	Total questions in quiz (5, 10, or 15)
Percentage Score	tblAiQuizzes	percentage	Number	-	Calculated score percentage (0 - 100)
Completion Timestamp	tblAiQuizzes	timestamp	Number	-	Epoch timestamp of quiz completion

tblAnnouncements (Announcements Collection)

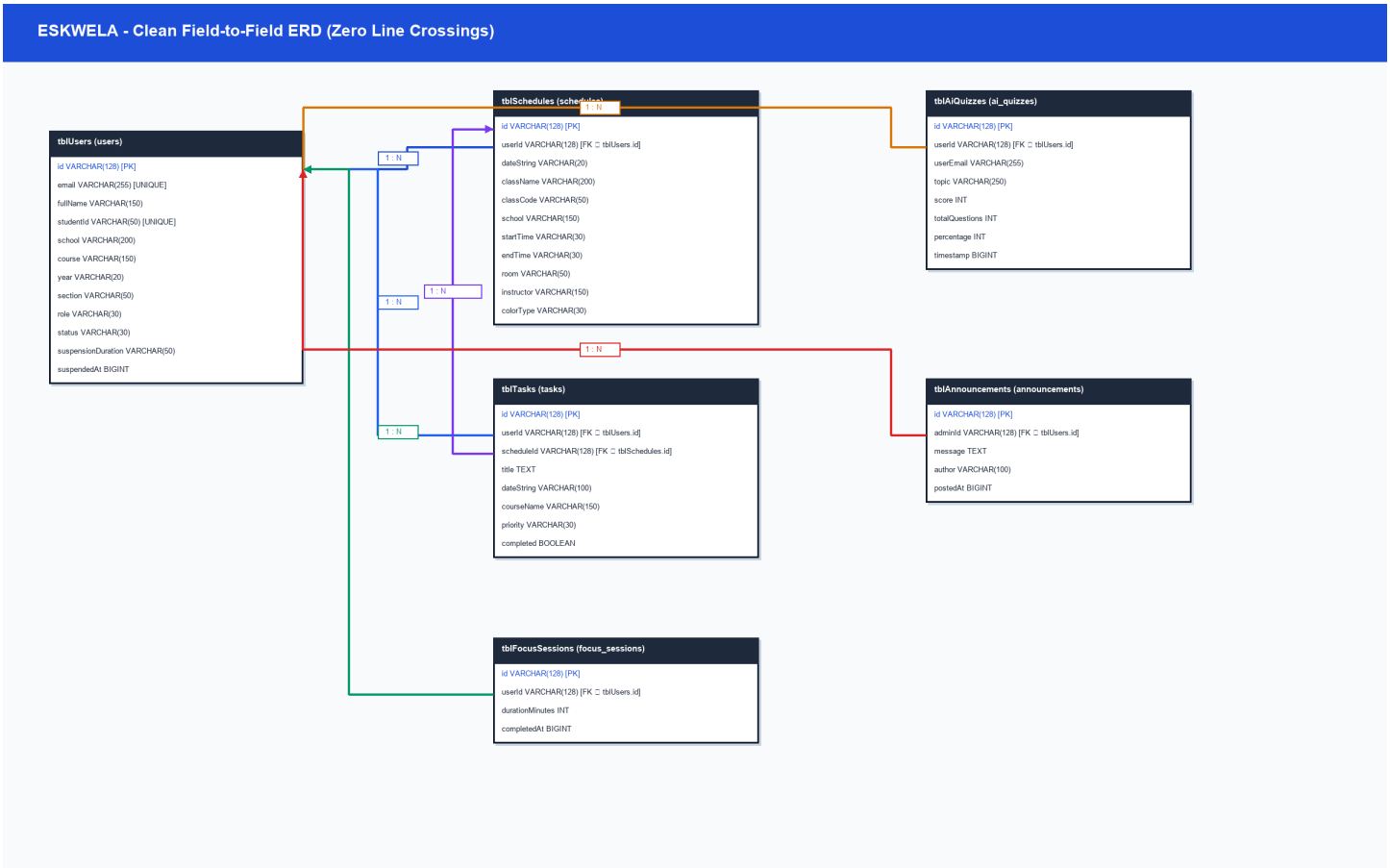
Field Name	Table	Database Field Name	Data Type	Length/Decimal Places	Description
Announcement ID	tblAnnouncements	id	Text	128	Unique announcement Primary Key
Admin Author ID	tblAnnouncements	adminId	Text	128	Foreign Key referencing tblUsers(id)
Announcement Message	tblAnnouncements	message	Text	1000	Official broadcast message text
Author Tag	tblAnnouncements	author	Text	100	Author credential tag (e.g. ADMIN)
Posted Timestamp	tblAnnouncements	postedAt	Number	-	Epoch timestamp when broadcasted

tblFocusSessions (Focus Sessions Collection)

Field Name	Table	Database Field Name	Data Type	Length/Decimal Places	Description
Session ID	tblFocusSessions	id	Text	128	Unique focus session Primary Key
User ID	tblFocusSessions	userId	Text	128	Foreign Key referencing tblUsers(id)
Duration Minutes	tblFocusSessions	durationMinutes	Number	-	Focus duration completed (2, 5, 10, 15, 25, 45, 60)
Completed Timestamp	tblFocusSessions	completedAt	Number	-	Epoch timestamp of session completion

3. Precise Field-to-Field Visual Entity-Relationship Diagram (ERD)

Zero-crossing, clean diagram with precise orthogonal arrows pointing directly from Foreign Keys [FK] to Primary Keys [PK]:



4. Visual System Use Case Diagram (Admin Access Scoped)

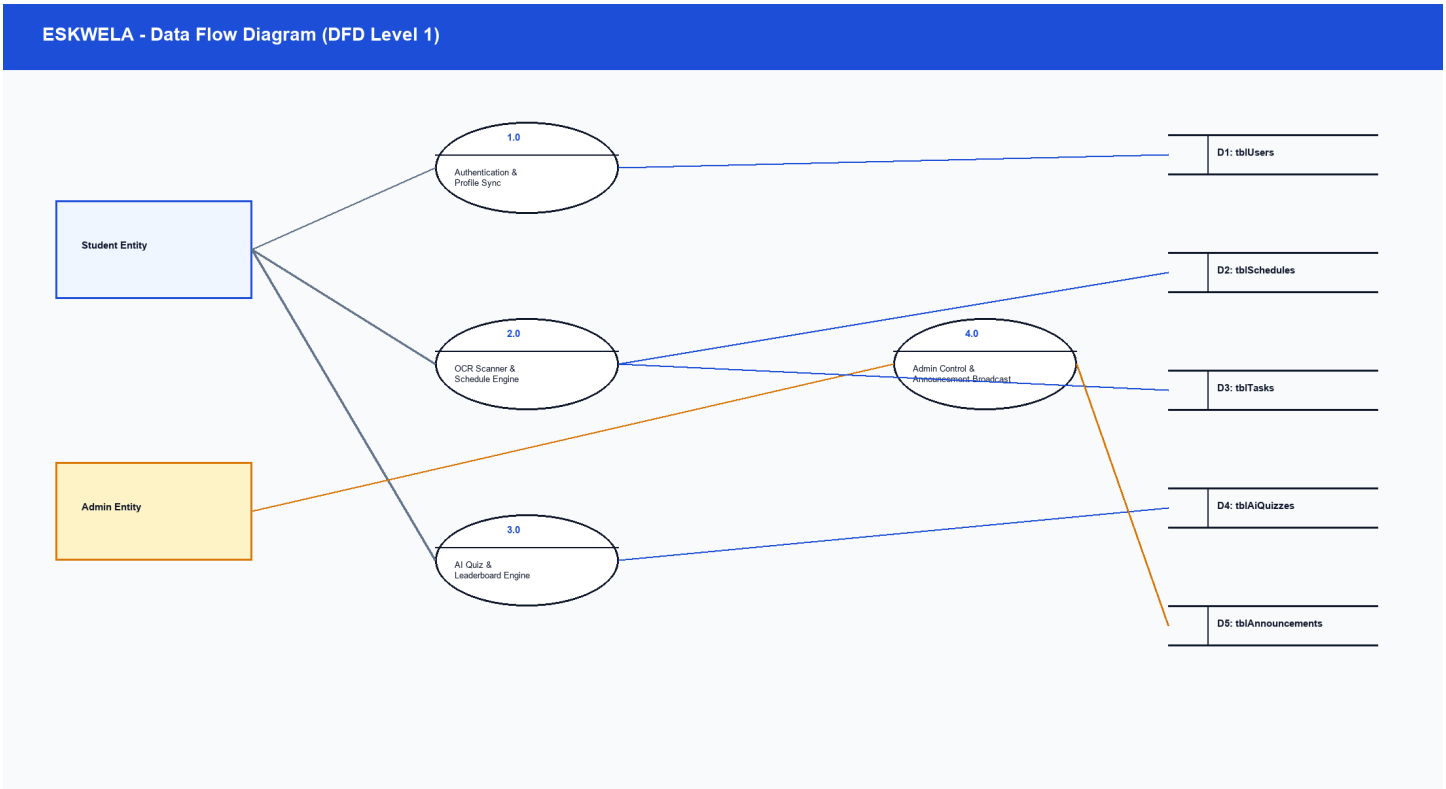
High-resolution diagram showing Student use cases and Admin restricted access (Admin Control Center & Announcements ONLY):

ESKWELA - System Use Case Diagram (Role Access Control)



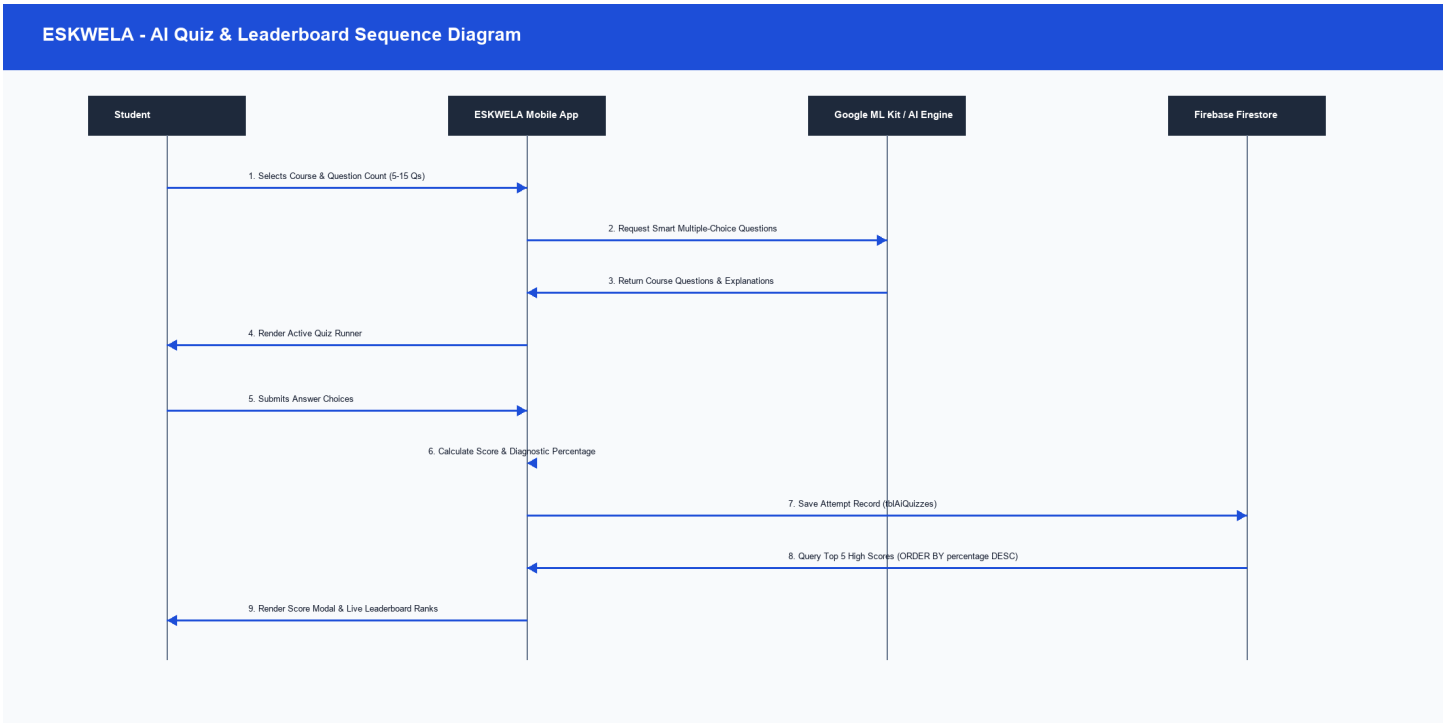
5. Visual Data Flow Diagram (DFD Level 1)

High-resolution diagram depicting data flow between entities, processes, and data stores:



6. Visual AI Quiz & Leaderboard Sequence Diagram

High-resolution diagram depicting the execution sequence for smart AI quiz generation and leaderboard rankings:



7. Database Script File (schema.sql)

Raw DDL creation script and sample seed data:

```
-- ========================================================================
-- ESKWELA DATABASE SCHEMA CREATION SCRIPT (DDL)
-- Backend Engine: Google Firebase Auth & Firestore NoSQL / Relational Export
-- ========================================================================
DROP TABLE IF EXISTS tblFocusSessions CASCADE;
DROP TABLE IF EXISTS tblAnnouncements CASCADE;
DROP TABLE IF EXISTS tblAiQuizzes CASCADE;
DROP TABLE IF EXISTS tblTasks CASCADE;
DROP TABLE IF EXISTS tblSchedules CASCADE;
DROP TABLE IF EXISTS tblUsers CASCADE;
CREATE TABLE tblUsers (
id VARCHAR(128) PRIMARY KEY,
email VARCHAR(255) NOT NULL UNIQUE,
fullName VARCHAR(150),
studentId VARCHAR(50) UNIQUE,
school VARCHAR(200),
course VARCHAR(150),
year VARCHAR(20),
section VARCHAR(50),
role VARCHAR(30) NOT NULL DEFAULT 'student',
status VARCHAR(30) NOT NULL DEFAULT 'active',
suspensionDuration VARCHAR(50),
suspendedAt BIGINT
);
CREATE TABLE tblSchedules (
id VARCHAR(128) PRIMARY KEY,
userId VARCHAR(128) NOT NULL,
dateString VARCHAR(20) NOT NULL,
className VARCHAR(200) NOT NULL,
classCode VARCHAR(50),
school VARCHAR(150),
startTime VARCHAR(30) NOT NULL,
endTime VARCHAR(30) NOT NULL,
room VARCHAR(50),
instructor VARCHAR(150),
colorType VARCHAR(30) NOT NULL,
FOREIGN KEY (userId) REFERENCES tblUsers(id) ON DELETE CASCADE
);
CREATE TABLE tblTasks (
id VARCHAR(128) PRIMARY KEY,
userId VARCHAR(128) NOT NULL,
scheduleId VARCHAR(128),
title TEXT NOT NULL,
dateString VARCHAR(100) NOT NULL,
courseName VARCHAR(150),
priority VARCHAR(30) NOT NULL DEFAULT 'MEDIUM',
completed BOOLEAN NOT NULL DEFAULT FALSE,
FOREIGN KEY (userId) REFERENCES tblUsers(id) ON DELETE CASCADE,
FOREIGN KEY (scheduleId) REFERENCES tblSchedules(id) ON DELETE SET NULL
);
CREATE TABLE tblAiQuizzes (
id VARCHAR(128) PRIMARY KEY,
userId VARCHAR(128) NOT NULL,
userEmail VARCHAR(255),
topic VARCHAR(250) NOT NULL,
score INT NOT NULL,
totalQuestions INT NOT NULL,
percentage INT NOT NULL,
timestamp BIGINT NOT NULL,
FOREIGN KEY (userId) REFERENCES tblUsers(id) ON DELETE CASCADE
);
CREATE TABLE tblAnnouncements (
id VARCHAR(128) PRIMARY KEY,
adminId VARCHAR(128) NOT NULL,
message TEXT NOT NULL,
author VARCHAR(100) NOT NULL DEFAULT 'ADMIN',
postedAt BIGINT NOT NULL,
FOREIGN KEY (adminId) REFERENCES tblUsers(id) ON DELETE CASCADE
);
CREATE TABLE tblFocusSessions (
id VARCHAR(128) PRIMARY KEY,
userId VARCHAR(128) NOT NULL,
durationMinutes INT NOT NULL,
```

```
completedAt BIGINT NOT NULL,
FOREIGN KEY (userId) REFERENCES tblUsers(id) ON DELETE CASCADE
);
-- SAMPLE SEED DATA
INSERT INTO tblUsers (id, email, fullName, studentId, school, course, year, section, role, status)
VALUES ('admin-uid-001', 'admin@eskwela.com', 'System Administrator', 'ADM-0001', 'ESKWELA University',
'Administration', 'Admin', 'ADMIN-01', 'admin', 'active');
INSERT INTO tblUsers (id, email, fullName, studentId, school, course, year, section, role, status)
VALUES ('student-uid-101', 'student@eskwela.com', 'Juan Dela Cruz', 'STU-4921', 'University of the Philippines', 'BS
Information Technology', '4th Year', 'IT41C1', 'student', 'active');
INSERT INTO tblSchedules (id, userId, dateString, className, classCode, school, startTime, endTime, room, instructor,
colorType)
VALUES
('sch-01', 'student-uid-101', '2025-05-12', 'CAPSTONE PROJECT 1', 'IT6398', 'University', '3:00 PM', '4:30 PM',
'301', '3 Units · Sec IT41C1', 'BLUE'),
('sch-02', 'student-uid-101', '2025-05-12', 'LOAD TESTING', 'CS6303', 'University', '12:00 PM', '2:30 PM', 'SLAB5',
'3 Units · Sec IT41C2', 'ORANGE'),
('sch-03', 'student-uid-101', '2025-05-12', 'SOFTWARE ENGINEERING', 'CS6209', 'University', '4:30 PM', '5:30 PM',
'204', '3 Units · Sec IT41C6', 'GREEN');
```